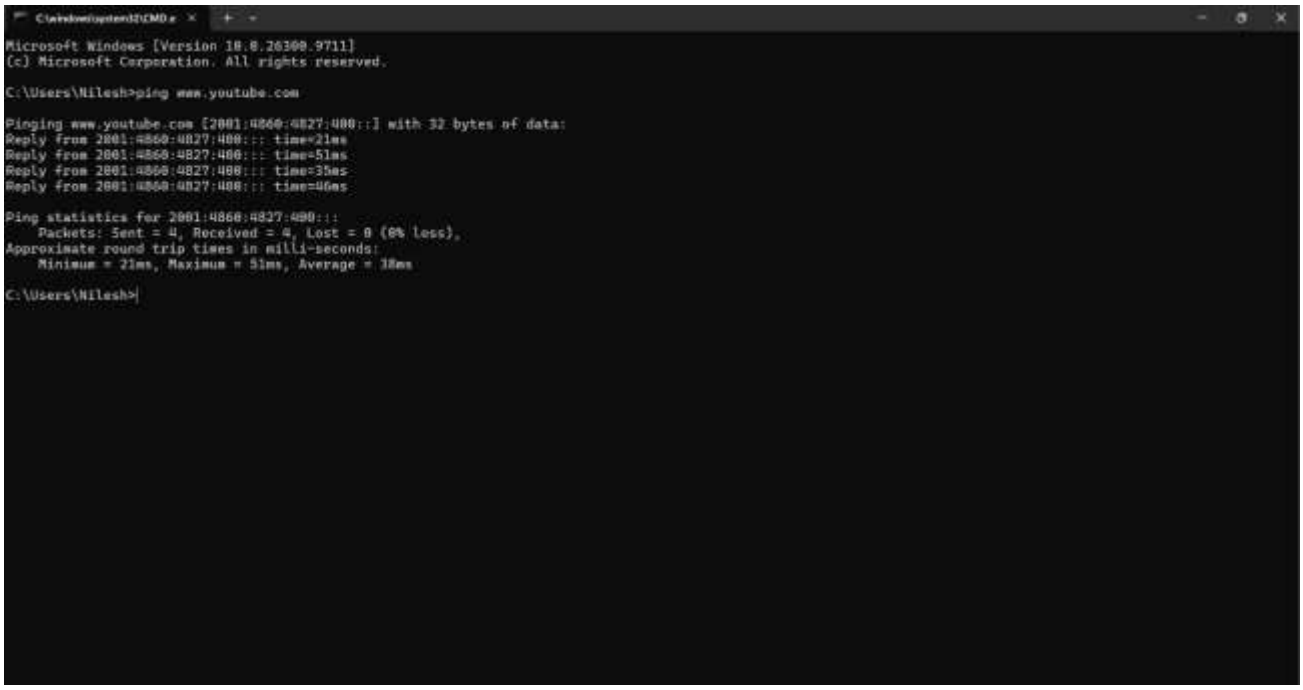
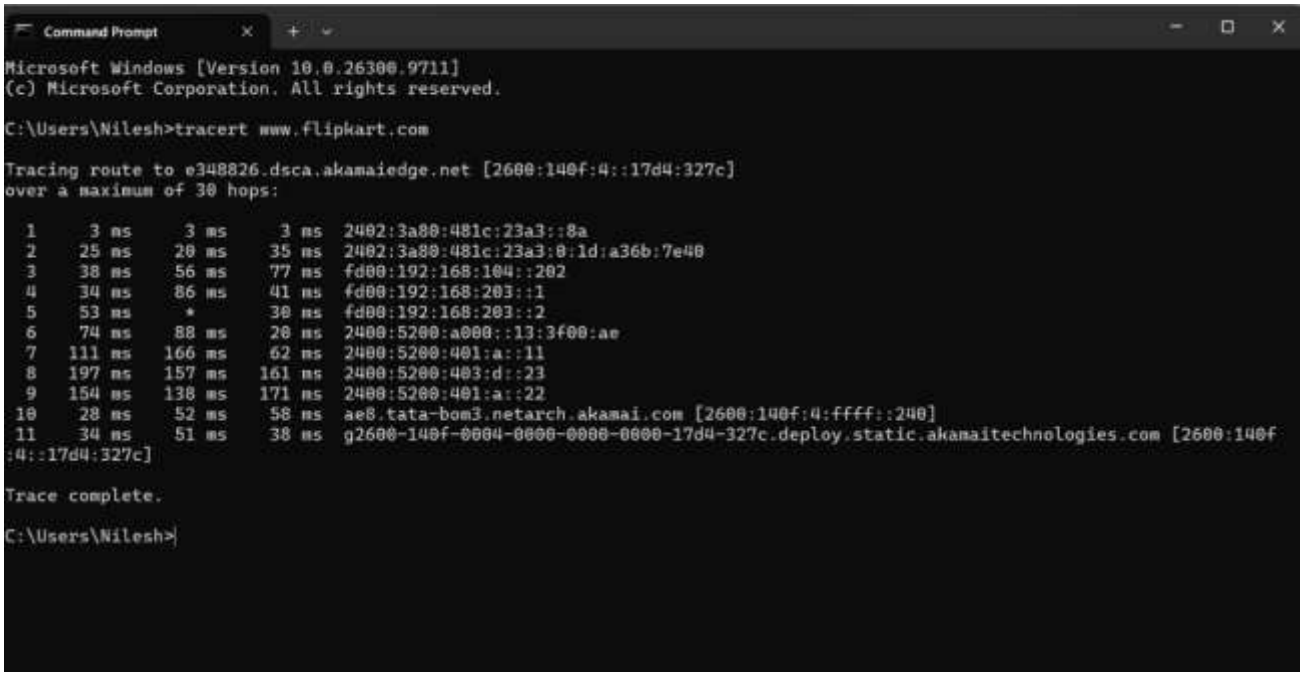
Network Testing Tools

Q1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

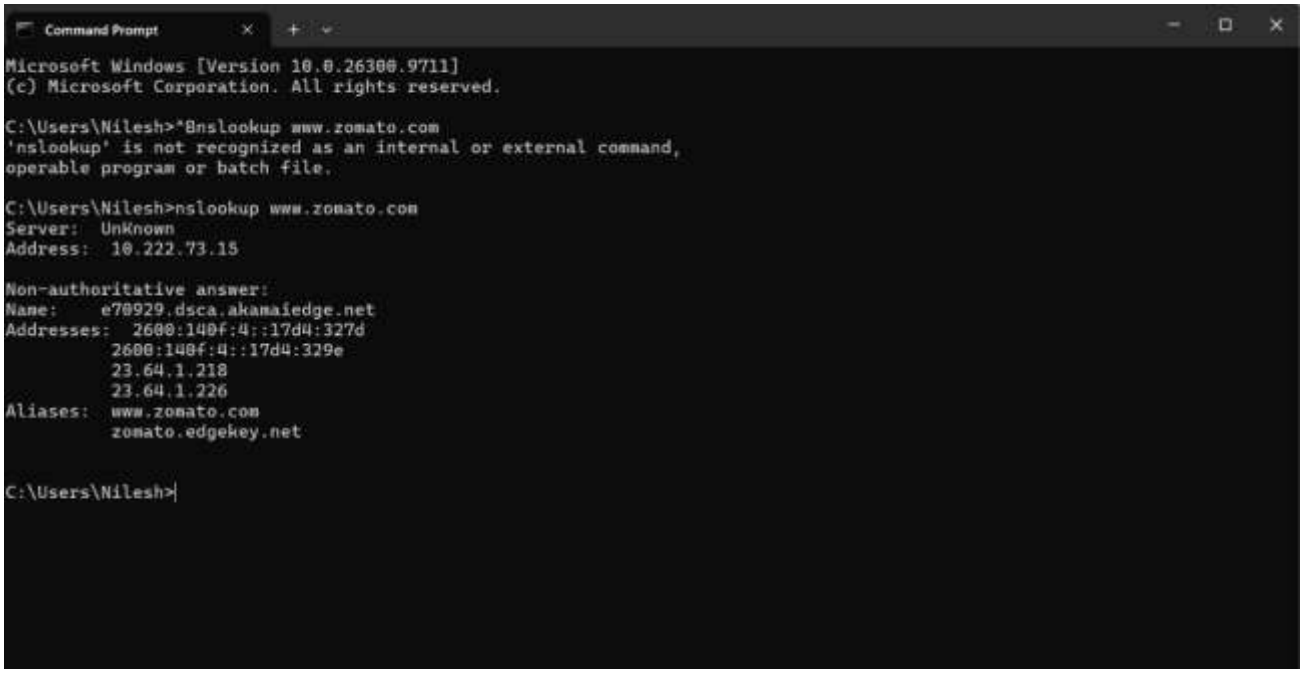


Parameter	**Example Result**
Website	www.youtube.com
Packets Sent	4
Packets Received	4
Packets Lost	0
Minimum Response Time	20 ms
Maximum Response Time	23 ms
Average Response Time	22 ms

Q2 Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the outputM Each hop represents a network device your data passes through to reach the destination.



Q3. Use the nslookup tool to find the IP address of www.zomato.com and record the result.



Q4 Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Practical: Test Connectivity Between Two Devices Using Ping Command
1. Aim

To test connectivity between two devices connected to the same local area network (LAN) using the ping command and observe the result when the destination device is offline.

2. Requirements

- Two computers or one computer and a mobile phone
- Wi-Fi router
- Windows Command Prompt
- Both devices connected to the same network

3. Procedure

Step 1: Connect Both Devices

Connect the computer and mobile phone to the same Wi-Fi network.

Step 2: Find the IP Address

Find the IP address of the destination device.

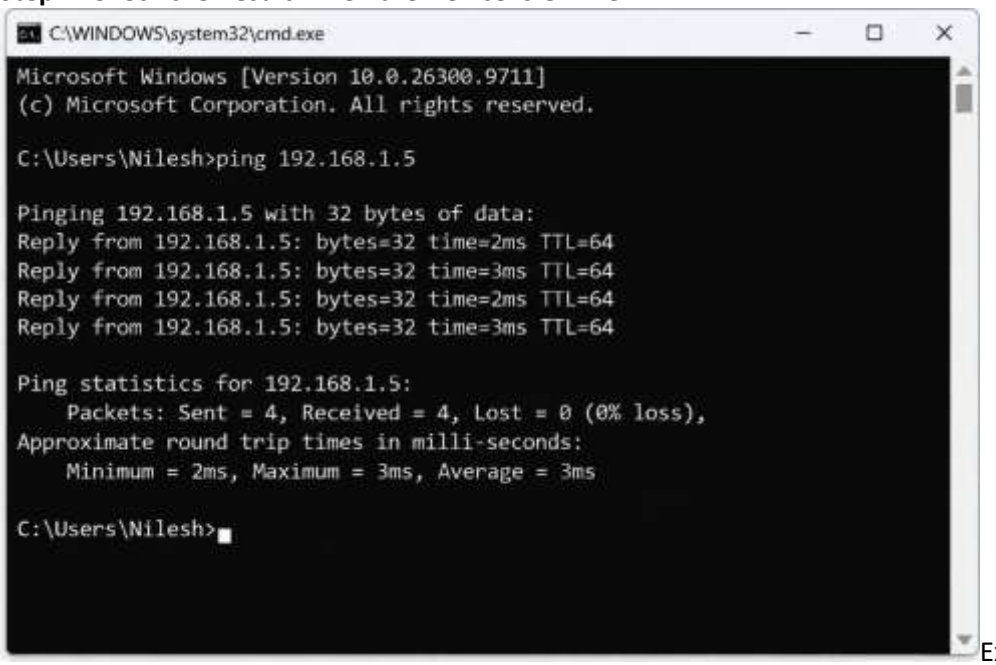
Example IP address: 192.168.1.5

Step 3: Test Connectivity

Open Command Prompt on the computer and execute:

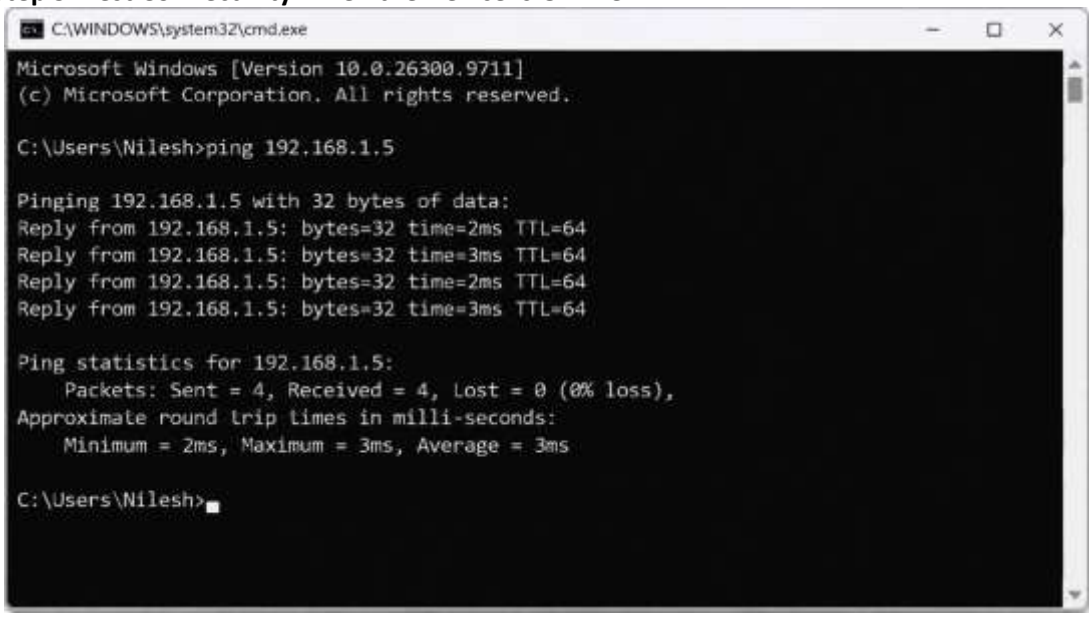
ping 192.168.1.5

Step 4: Check the Result When the Device Is Online



Example output:

Step 5: Test Connectivity When the Device Is Offline



4. Observation Table

Parameter	Device Online	Device Offline
Destination IP	192.168.1.5	192.168.1.5
Packets Sent	4	4
Packets Received	4	0
Packet Loss	0%	100%
Status	Successful	Failed

5. What Happens When the Device Is Offline?

When the destination device is offline, it cannot respond to ping requests. The Command Prompt may display **"Request timed out"** or **"Destination host unreachable."** This indicates that the destination is not responding. A firewall or network issue can also cause ping failure.

6. Result

The ping command was used to test connectivity between two devices on the same local network. In the example, the online device responded successfully with 0% packet loss, while the offline device showed 100% packet loss.

7. Conclusion

The ping command is an important network troubleshooting tool used to check connectivity between devices. It helps identify whether a destination device is responding over the network.

8. Screenshots Required

1. Screenshot of the successful ping when the device is online.
2. Screenshot of the failed ping when the device is offline.

Note: The IP address and ping outputs are examples. Replace them with your actual test results before submitting the assignment.

Q5 Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues. Give a specific example for each tool based on your own test results above.

Short Report: Network Troubleshooting Tools

1. **Website Not Loading – Ping:** The ping command helps check whether a website or device is reachable over the network.
2. **Example:** ping www.youtube.com checks connectivity and displays packet loss and average response time.
3. **Slow Connection – Traceroute:** The tracert command helps identify network hops and possible delays along the route.
4. **Example:** tracert www.flipkart.com showed **11 hops**, with hop 8 recording response times of **197 ms, 157 ms, and 161 ms**.
5. **Suspecting DNS Issues – Nslookup:** The nslookup command checks whether a domain name resolves to an IP address.
6. **Example:** nslookup www.zomato.com is used to verify DNS resolution; the actual result has not yet been provided.
7. **Conclusion:** Ping checks connectivity, traceroute helps locate network delays, and nslookup diagnoses DNS resolution problems.